SOMAIYA VIDYAVIHAR UNIVERSITY

16.05.2024 (E)

Semester: January 2024 – April 2024 Maximum Marks: 100 Examination: ESE Examination Duration:3 Hrs.		
Programme code: 66 Programme: Honour Programme in Artificial Intelligence	Class: TY (B.Tech.)	Semester: VI (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Information Technology	
Course Code: 116h66C601	Name of the Course: Deep Learning	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	List different challenges which have motivated Deep Learning.	5
ii)	Explain Probability Conditional Probabilities, Posterior Probability with examples.	5
iii)	Describe various resampling methods.	5
iv)	What is Stochastic Gradient Descent? What are its drawbacks?	5
v)	What is the need for activation function? Describe any one most widely used activation function in deep learning.	5
vi)	Describe any five parameters used to evaluate the performance of deep learning models.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	What are hyperparameters? What is the difference between parameter tuning and hyperparameter tuning? What are the advantages of hyperparameter tuning?	5
ii)	Describe any one optimization algorithms with Adaptive Learning Rate.	5
	OR	
Q2 A	What is regularization? What are the benefits of regularization? Elaborate any two regularization techniques.	10
Q 2 B	Solve any **One**	**10**
i)	Explain any four loss functions with formulas.	10
ii)	What are the four major architectures of deep networks? Describe any two with diagram.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
	a. Consider and input of size 28 X 28 X 192. Apply a 1 X 1 convolution using 32 filters. What will be size of the output? What is effect of using 1 X 1 convolution. b. Consider the input image data as	10

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ii)	Explain the following terms related to CNN: a. Transfer learning b. Padding c. Pooling d. Flattening Layer e. Fully-Connected Layer	10
iii)	How do CNNs deal with overfitting? Describe any four techniques to mitigate overfitting in CNNs.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	What is a Gated Recurrent Unit (GRU), and how does it differ from an LSTM cell?	10
ii)	What are Recurrent Neural Networks (RNNs), and how do they differ from feedforward neural networks? Describe how the hidden state in an RNN operates?	10
iii)	What are Long Short-Term Memory (LSTM) networks, and how do they address the vanishing gradient problem?	10

Que. No.	Question	Max. Marks
Q5	Write short on any **four**	**20**
i)	Unsupervised Pretrained Networks	5
ii)	Deep Belief Networks	5
iii)	ReLU and Leaky ReLU activation function	5
iv)	Variants of basic convolution function	5
v)	Recursive Neural Networks	5
vi)	Sequence-to-Sequence Architectures	5